

Single-cell foundation models represent genes in context, but reveal no context-specific gene function beyond co-expression

Ihor Kendiukhov^{a,*}

^a Institute of Medical Genetics and Applied Genomics, University of Tübingen, Calwerstraße 7, 72076 Tübingen, Germany

* Corresponding author: Ihor Kendiukhov, Institute of Medical Genetics and Applied Genomics, University of Tübingen, Calwerstraße 7, 72076 Tübingen, Germany. E-mail: kendiukhov@gmail.com

Short title: Contextual gene representations in single-cell models

Keywords: Single-cell foundation models; Contextual gene representations; Mechanistic interpretability; Gene co-expression; Gene embeddings; Transformers; Model evaluation

Word count of the abstract: 249